Rotation and Composition Findings

12 February 2026

Questions

- Is the composed kernel rotation invariant?
- Why is composite accuracy lower than regular all-kernel model?

Rotation-Invariance

- Composition is a weighted sum: $K_{comp} = \sum_i w_i K_i$.
- This is not automatically rotation invariant.
- Non-rotation-invariant kernel families in the 9-bank: anisotropic Gaussian, Gabor, HOG, LBP, GLCM.
- For the analyzed composition run, $90/180/270^\circ$ checks show large differences.
- **Current composite kernel is not rotation invariant. But...**

Non-Invariant But Still Robust

- $90^\circ/180^\circ$ test rotation still gave stable accuracy:
- Regular model: ΔAUC $+0.0017$, ΔACC $+0.0123$
- Composite model: ΔAUC $+0.0086$, ΔACC $+0.0241$
- Takeaway: non-invariant kernels can still yield stable accuracy.

Why Composite is Lower

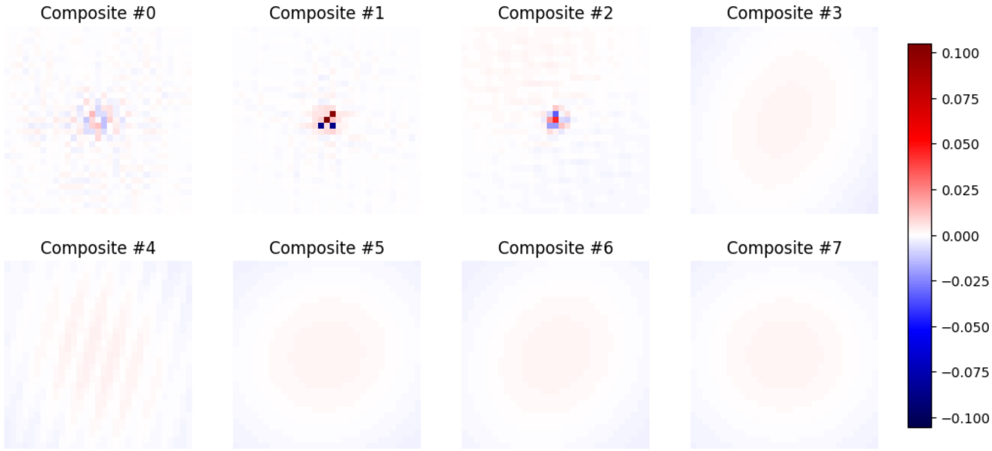
- Feature collapse $\rightarrow$ fix: use a small bank of composites (e.g., 8–32) instead of one.
- Nonlinearity mismatch with `abs_max` $\rightarrow$ fix: make the response linear by replacing `abs_max` with linear pooling (mean or sum), so composing kernels before response is closer to combining features after response.

Best Current Run

Model	Test AUC	Test ACC
Regular (separate kernels)	0.9376	0.8888
Composite (8 kernels)	0.9026	0.8299

composition remains below separate kernels in this run.

Composite Kernels (Grid)



Current Pipeline and Next Step

- Current: Kernel generation is random.
- Change: Smarter sampling using QMC or Latin Hypercube.
- Current: normalization is applied to composed kernels; non-composed kernels are used without an extra normalization stage. Prior runs without extra normalization performed better.
- Change: test per-family normalization (different normalization per kernel type).

Conclusion

- Current composite is not rotation invariant.
- Composite underperforms regular model mainly due to information compression and nonlinearity effects.
- Matching regular accuracy is more realistic with multiple composites than a single one.
- Larger composition pools can look “zoomed” due to cancellation and normalization.